Block encodings for numerical linear algebraists

How the combination rules are realized
(supplement to "Preconditioning for quantum linear solvers in PDE-constrained optimization")

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Abstract

Section 2 of the paper lists how block encodings combine: products, Kronecker products, linear combinations, stacking, concatenation and sparse matrices, each with its subnormalization. This supplement shows, in pure matrix terms, how each rule is realized as an explicit unitary matrix, and why the subnormalizations are what they are. Every construction was built as a dense matrix and checked numerically, with the script `qp_block_encoding_rules.py` in the folder `preconditioning/code` (Table 3).

1 Conventions

Registers are Kronecker factors. A quantum computer with q qubits acts on $\mathbb{C}^2 \otimes \cdots \otimes \mathbb{C}^2 = \mathbb{C}^{2^q}$. A *register* is a group of these factors. A *circuit* is a product of unitary matrices, each acting on a few factors and as the identity on the rest. Two operations are free:

- permuting tensor factors, which is only a relabelling of qubits;
- adding fresh factors that start in the first basis vector e_0.

Ancillas. The qubits are split into two groups. The *system register* holds the data: a vector $v \in \mathbb{C}^n$ with $n = 2^k$ uses k qubits. All further qubits are *ancillas* (auxiliary qubits), a of them, so the full space is $\mathbb{C}^{2^a} \otimes \mathbb{C}^n$. An ancilla is defined by its role, not by any physical difference: it is an extra tensor factor that

- starts in the first basis vector e_0 (in physics notation $|0\rangle$), so that the input v becomes $e_0 \otimes v$;
- serves as workspace for a computation that could not be done on the system register alone, because only unitary matrices can be applied;
- at the end is either *measured* (see the paragraph on postselection below) or returned to e_0, so that it is no longer coupled to the system and can be reused.

In matrix terms, adding a ancillas embeds $\mathbb{C}^n$ into the larger space $\mathbb{C}^{2^a n}$ as the first block, $v \mapsto [v; 0; \ldots; 0]$. A unitary on the larger space can act on that block in a non-unitary way, as long as it moves the surplus into the other blocks. This is exactly what a block encoding does: its leading block, "ancillas in e_0 on input and on output", is A/α, and everything else is where the remainder goes.

Ancillas get names according to their role:

- a *flag* qubit marks success (e_0) or failure (e_1), as in a rotation with amplitude $g(\mu_j)$ on e_0 (Section 13);

- an *index register* of $\lceil \log_2 k \rceil$ qubits holds a label $j = 1, \ldots, k$, e.g. which term of a linear combination or which level of a multilevel factor is applied;
- *work registers* hold intermediate results of reversible arithmetic, e.g. a rotation angle, and must be returned to e_0 ("uncomputed") afterwards.

The number of ancillas is part of the cost (qubits), but usually a minor one: one for the Halmos dilation, $\lceil \log_2 k \rceil$ for a linear combination of k terms, $a_A + a_B$ for a product or a Kronecker product. The more important point is that ancillas must be *clean*: a block encoding requires them to be in e_0 on input, and reusing an ancilla that is not back in e_0 brings discarded components into the leading block (Section 4).

Ordering. We write extra registers (ancillas, index registers) as the *outer* Kronecker factors and the system as the innermost one. A vector in $\mathbb{C}^2 \otimes \mathbb{C}^n$ is therefore a stacked vector $[v_0; v_1]$, and $e_0 \otimes v = [v; 0]$.

States and measurement. The state of a quantum computer is a *unit* vector ψ; a vector v is stored as the state $v/\|v\|$. A circuit maps ψ to $U\psi$, which is again a unit vector because U is unitary. Measuring a register is the only way to obtain information, and in matrix terms it works as follows (the Born rule). Split ψ into blocks according to the basis states of the measured register, $\psi = [\psi_0; \psi_1; \ldots]$, with Π_j the orthogonal projector onto block j. Then

- outcome j occurs with probability $\|\Pi_j \psi\|^2 = \|\psi_j\|^2$; these probabilities sum to $\|\psi\|^2 = 1$;
- after outcome j the state is $\Pi_j \psi / \|\Pi_j \psi\|$, i.e. the block ψ_j, renormalized, with all other blocks set to zero.

A measurement thus picks one block at random, with probability equal to its squared norm, and renormalizes it. The norm of the selected block is lost; only its direction survives.

Block encodings. A unitary $U \in \mathbb{C}^{2^a n \times 2^a n}$ is a block encoding of $A \in \mathbb{C}^{n \times n}$ with subnormalization α if its leading block is A/α:

$$(e_0^T \otimes I_n)\, U\, (e_0 \otimes I_n) = A/\alpha, \qquad \text{i.e.} \qquad U = \begin{bmatrix} A/\alpha & * \\ * & * \end{bmatrix}.$$

Applying U to $[v; 0]$ gives $[Av/\alpha;\ w]$ for some w. Measuring the ancilla and keeping the outcome "0" is called *postselection*.

What postselection yields, and what α costs. The input state is the unit vector $u = [v/\|v\|;\, 0]$, and after the circuit

$$\psi = Uu = \begin{bmatrix} Av/(\alpha\|v\|) \\ w' \end{bmatrix}.$$

By the measurement rule, the outcome "ancilla = 0" occurs with probability equal to the squared norm of the first block,

$$p = \frac{\|Av\|^2}{\alpha^2 \|v\|^2},$$

and the state afterwards is that block, renormalized:

$$\frac{Av/(\alpha\|v\|)}{\|Av\|/(\alpha\|v\|)} = \frac{Av}{\|Av\|}.$$

Both α and $\|v\|$ cancel. Hence:

- the *result* of postselection does not depend on α: it is always the direction $Av/\|Av\|$;
- the *success probability* does: it scales like $1/\alpha^2$. A loose encoding ($\alpha \gg \|A\|$) means more failed runs, about $1/p$ of them, or about $1/\sqrt{p}$ rounds of amplitude amplification (Section 12).

This is why α is a cost and not part of the answer. Note also that α is a property of the encoding, fixed by $\|A\|$ and the tightness of the circuit, while $\|Av\|$ depends on the input: even with the ideal $\alpha = \|A\|$, p equals one only if v is a top right singular vector of A, and it is about $\sigma_{\min}^2/\alpha^2$ if v is close to the smallest one. Instead of measuring, one can also continue to apply further unitaries to ψ; this is how QSVT composes block encodings without intermediate postselection.

For rectangular $A \in \mathbb{C}^{m\times n}$ the rows and the columns are selected by different projections. These are P_{out} and P_{in}, both with orthonormal columns, and we require $P_{\text{out}}^* U P_{\text{in}} = A/\alpha$.

Why $\alpha \geq \|A\|$. Every block of a unitary matrix has norm at most one, so $\|A/\alpha\| \leq 1$. The subnormalization is the price of embedding a non-unitary matrix in a unitary one. It enters the cost of everything downstream: success probabilities, and the degree of QSVT polynomials.

2 A single matrix: the unitary dilation

Lemma 1 (Halmos dilation)**.** *Let $\|X\| \leq 1$. Then*

$$U = \begin{bmatrix} X & (I - XX^*)^{1/2} \\ (I - X^*X)^{1/2} & -X^* \end{bmatrix}$$

is unitary.

Proof. Use $X^* f(XX^*) = f(X^*X)X^*$ for $f(t) = (1-t)^{1/2}$. This makes the off-diagonal blocks of U^*U vanish, and

$$U^*U = \begin{bmatrix} X^*X + (I - X^*X) & 0 \\ 0 & (I - XX^*) + XX^* \end{bmatrix} = I.$$

□

So every A has a block encoding with $\alpha = \|A\|$ and one ancilla qubit. The dilation is dense, however, and has no short circuit in general. The point of the rules below is to build block encodings of *structured* matrices from small, cheap pieces. The price is a subnormalization that may exceed $\|A\|$.

3 The ideal subnormalization and what circuits achieve

Definition. The *ideal subnormalization* of A is $\alpha = \|A\|$. It is a lower bound for every block encoding, and the Halmos dilation attains it. The dilation, however, is dense and has no efficient circuit in general. Efficient block encodings are assembled from structured pieces by the rules of the following sections. At each step the subnormalization can exceed the norm. The table below summarizes when a rule is tight, i.e. when its α equals the norm of the result.

rule	α versus norm	tight when
AB	$\alpha_A\alpha_B \geq \|A\|\|B\| \geq \|AB\|$	A is maximal on the range of B's top singular vector
$A \otimes B$	$\alpha_A\alpha_B$ vs. $\|A \otimes B\| = \|A\|\|B\|$	always (if α_A, α_B are tight)
$\sum_j c_j A_j$	$\sum_j \|c_j\|\alpha_j \geq \|\sum_j c_j A_j\|$	all terms maximal on a common vector, phases aligned
$[A_0; \ldots; A_{k-1}]$	$(\sum_j \alpha_j^2)^{1/2}$ vs. $\|\sum_j A_j^* A_j\|^{1/2}$	all A_j maximal on a common input vector
$[A_0, \ldots, A_{k-1}]$	$(\sum_j \alpha_j^2)^{1/2}$ vs. $\|\sum_j A_j A_j^*\|^{1/2}$	all A_j maximal on a common output vector
sparse A	$a_{\max}\sqrt{s_r s_c}$ (Schur-test bound) vs. $\|A\|$	rarely

The common mechanism. A loss occurs when the pieces attain their norms on *different* vectors. In PDE discretizations the typical conflict is between oscillatory vectors, which maximize stiffness and gradient operators, and smooth vectors, which maximize mass and prolongation operators.

Examples.

- *A tight LCU.* $I - S$ with the cyclic shift S has $\alpha = 2 = \|I - S\|$ for even n: both terms are maximal on the alternating vector, with aligned signs.
- *An LCU that loses a factor* 2. The 2D stiffness matrix $K = K_1 \otimes M_1 + M_1 \otimes K_1$ as an LCU of two Kronecker products has $\alpha = 2\|K_1\|\|M_1\| \to 8$, while $\|K\| \to 4$ (Q_1, $h = 2^{-L}$; the ratio is 1.993, 2.000, 2.000, 2.000 for $L = 4, 6, 8, 10$). $\|K_1\|$ is attained by the most oscillatory sine vector, $\|M_1\|$ by the smoothest, and no single vector makes both terms maximal.
- *A product that loses* $2^{L-\ell}$. $\|F_1\|\|P_\ell\|/\|F_1 P_\ell\|$ (gradient factor times prolongation) equals $2^{L-\ell}$ up to the coarsest levels (paper, Table 4), for the same reason. The commuting lemma of the paper removes this loss by applying the gradient on the coarse level first.
- *Concatenation over levels loses* $\sqrt{L}$. For $G = [T_1 G_1, \ldots, T_L G_L]$ the rule gives $\alpha \approx \sqrt{L}\max_\ell \|G_\ell\|$. Multilevel theory (a strengthened Cauchy–Schwarz inequality between levels) keeps $\|G\| = O(1)$, because different levels mostly act on different frequency bands. Their output ranges are far from aligned, so the rule is not tight.
- *Sparse access loses a factor 3* in the random example of Table 3: $\alpha = 15.3$ against $\|A\| = 5.1$.

What this means for the paper. The preconditioned KKT matrix is encoded as $\mathcal{B} = \mathcal{G}^T \mathcal{J} \mathcal{G}$ with $\mathcal{G} = I_2 \otimes G$, where $G = [F_{\text{val}} R; \beta^{1/4} F_\nabla R]$ maps multilevel coefficients to values and gradients at quadrature points. The reflection $\mathcal{J}$ has $\alpha = 1$, so $\alpha_\mathcal{B} = \alpha_G^2$.

- *Ideal case:* $\alpha_G = \|G\|$, giving $\alpha_\mathcal{B} = \|G\|^2 = \|R^T A_\beta R\| \leq c_1$. The paper's Table 1 has $\|G\| \approx 1.7$–2.7, hence $\kappa_\alpha(\mathcal{B}) \approx 5$–$10$.
- *Even the ideal case is above* $\|\mathcal{B}\|$: $\|\mathcal{B}\| \leq \|G\|^2$ since $\|\mathcal{J}\| = 1$. This is the price of writing an indefinite matrix with a reflection in the middle. Here both quantities are $O(1)$.
- *Actual construction:* $\|G\| \leq \alpha_G \leq C_d\sqrt{L}$, with losses from three places: the 1D LCUs, the stack of values and gradients, and the concatenation over levels. The pessimistic sparse-access constant is $C_3 \approx 45$.

- *Why measuring α_G matters:* it enters the cost squared. For $d = 3$ the gap between the bounds is up to $(45\sqrt{L}/2)^2 \approx 500L$. Building the encoding explicitly and measuring α_G (to-do item 1.1) decides where between the two bounds we are.

4 Products: fresh ancillas

Let $U_A = \begin{bmatrix} X_A & Y_A \\ Z_A & * \end{bmatrix}$ and $U_B = \begin{bmatrix} X_B & Y_B \\ Z_B & * \end{bmatrix}$ be block encodings with $X_A = A/\alpha_A$ and $X_B = B/\alpha_B$.

The naive product fails. If both use the *same* ancilla,

$$U_A U_B = \begin{bmatrix} X_A X_B + Y_A Z_B & * \\ * & * \end{bmatrix}.$$

The "garbage" $Z_B v$ that U_B moves into the ancilla-1 block is brought back into the leading block by Y_A.

The construction. Give each factor its own ancilla. Order the space as (ancilla of A) $\otimes$ (ancilla of B) $\otimes$ system. Let $\widetilde{U}_A$ act as U_A on the ancilla of A and the system, and as the identity on the ancilla of B; define $\widetilde{U}_B$ in the same way. Then

$$\left(e_0 \otimes e_0 \otimes I\right)^T \widetilde{U}_A \widetilde{U}_B \left(e_0 \otimes e_0 \otimes I\right) = X_A X_B = \frac{AB}{\alpha_A \alpha_B}.$$

The reason is that $\widetilde{U}_B$ puts its garbage into the block where the ancilla of B equals 1. $\widetilde{U}_A$ never touches that ancilla, so the garbage stays orthogonal to the leading block.

- *Cost:* one call to each factor, plus the extra ancillas.
- *Subnormalization:* $\alpha_A \alpha_B$, which can be much larger than $\|AB\|$. This is the mechanism behind the failure of the naive product encoding of $\widetilde{R}^T \mathcal{A} \widetilde{R}$ in the paper.

5 Kronecker products

$U_A \otimes U_B$ acts on (ancilla$_A$, system$_A$, ancilla$_B$, system$_B$). After the free permutation to (ancilla$_A$, ancilla$_B$, system$_A$, system$_B$), its leading block is

$$X_A \otimes X_B = \frac{A \otimes B}{\alpha_A \alpha_B}.$$

No calls are combined at all: the two circuits run side by side on disjoint qubits. This is why the Kronecker structure of Q_1 matrices on Cartesian grids is so valuable. A d-dimensional factor costs d one-dimensional circuits.

6 Linear combinations of unitaries (LCU)

Let $\sum_{j=0}^{k-1} c_j A_j$ be given, with block encodings U_j of A_j (subnormalization α_j), $k = 2^m$ terms, and $c_j > 0$. Complex phases of c_j are absorbed into U_j. Let $s = \sum_j c_j \alpha_j$.

- *PREPARE:* a unitary $V \in \mathbb{C}^{k \times k}$ with first column $Ve_0 = p$, where $p_j = \sqrt{c_j \alpha_j / s}$. It acts on an index register of m qubits.

- *SELECT:* the block-diagonal unitary SEL = blkdiag$(U_0, \dots, U_{k-1})$, i.e. "apply U_j if the index register is e_j".

Then

$$W = (V^* \otimes I)\,\mathrm{SEL}\,(V \otimes I)$$

has the leading block (index 0, ancilla 0)

$$\sum_j \bar{p}_j\, p_j\, \frac{A_j}{\alpha_j} = \sum_j \frac{c_j \alpha_j}{s} \frac{A_j}{\alpha_j} = \frac{1}{s} \sum_j c_j A_j.$$

So $\alpha = \sum_j c_j \alpha_j$.

Example: the difference matrix. Let S be the cyclic shift, a permutation matrix and hence unitary, with $\alpha = 1$. With $k = 2$, $p = \frac{1}{\sqrt{2}}(1,1)$, $V = H = \frac{1}{\sqrt{2}}\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$ and SEL $=$ blkdiag$(I, -S)$,

$$W = (H \otimes I) \begin{bmatrix} I & 0 \\ 0 & -S \end{bmatrix} (H \otimes I) = \frac{1}{2} \begin{bmatrix} I - S & I + S \\ I + S & I - S \end{bmatrix},$$

a block encoding of $I - S$ with $\alpha = 2$. The 1D quadrature factors F_0 and F_1 of the paper are built this way. They are two-term combinations of shifts with diagonal weights.

7 Stacking

The stack $\begin{bmatrix} A_0 \\ \vdots \\ A_{k-1} \end{bmatrix} \in \mathbb{C}^{kn \times n}$ is tall, and the index register becomes part of the *output*. Take V with $Ve_0 = p$, where $p_j = \alpha_j / N$ and $N = \left(\sum_j \alpha_j^2\right)^{1/2}$, and

$$W = \mathrm{SEL}\,(V \otimes I).$$

On an input $e_0 \otimes e_0 \otimes v$ (index 0, ancilla 0), V produces $\sum_j p_j\, e_j \otimes e_0 \otimes v$. SEL then applies U_j in branch j. Keeping ancilla 0 but *all* index values gives

$$\sum_j e_j \otimes \frac{p_j}{\alpha_j} A_j v = \frac{1}{N} \begin{bmatrix} A_0 v \\ \vdots \\ A_{k-1} v \end{bmatrix}.$$

This is a rectangular block encoding with P_{in}: (index 0, ancilla 0), and P_{out}: (any index, ancilla 0). Its subnormalization is $N = \left(\sum_j \alpha_j^2\right)^{1/2}$. In the paper this stacks the value rows and the d gradient rows of G_ℓ.

8 Concatenation

The concatenation $[A_0, \dots, A_{k-1}] \in \mathbb{C}^{n \times kn}$ is wide, and the index register is part of the *input*. With the same V,

$$W = (V^* \otimes I)\,\mathrm{SEL}.$$

An input $\sum_j e_j \otimes e_0 \otimes v_j$ goes to $\sum_j e_j \otimes (e_0 \otimes A_j v_j / \alpha_j + \dots)$. Applying V^* and keeping index 0 and ancilla 0 gives

$$\sum_j \bar{p}_j \frac{A_j v_j}{\alpha_j} = \frac{1}{N} \sum_j A_j v_j.$$

So $\alpha = N = \left(\sum_j \alpha_j^2\right)^{1/2} \le \sqrt{k}\max_j \alpha_j$. The factor $\sqrt{k}$ is the *price of summing*: the vectors $A_j v_j$ add up coherently only in the one index branch that is kept. In the paper this is the concatenation over the L levels, $G = [T_1 G_1, \ldots, T_L G_L]$. It is the source of $\alpha_G = O(\sqrt{L})$, and the reason the paper asks whether the $\sqrt{L}$ can be avoided.

9 Sparse matrices

Let $A \in \mathbb{C}^{n\times n}$ have at most s_r nonzeros per row and s_c per column, and let $a_{\max} = \max|a_{ij}|$. Work in $\mathbb{C}^n \otimes \mathbb{C}^n \otimes \mathbb{C}^2$ and define, for each row i and column j, the unit vectors

$$\psi_i = \frac{1}{\sqrt{s_r}} \sum_{j\in\mathcal{R}_i} e_i \otimes e_j \otimes \Big(\frac{\bar{a}_{ij}}{a_{\max}} e_0 + \sqrt{1 - \tfrac{|a_{ij}|^2}{a_{\max}^2}}\, e_1\Big), \qquad \phi_j = \frac{1}{\sqrt{s_c}} \sum_{i\in\mathcal{C}_j} e_i \otimes e_j \otimes e_0.$$

Here $\mathcal{R}_i$ ($\mathcal{C}_j$) is the nonzero pattern of row i (column j), padded with zero entries to exactly s_r (s_c) indices.

- The ψ_i are orthonormal, because they differ in the first factor. The ϕ_j are orthonormal, because they differ in the second.
- $\psi_i^* \phi_j = a_{ij}/(a_{\max}\sqrt{s_r s_c})$.

With the isometries $T_L = [\psi_1, \ldots, \psi_n]$ and $T_R = [\phi_1, \ldots, \phi_n]$,

$$\frac{A}{a_{\max}\sqrt{s_r s_c}} = T_L^* T_R.$$

Completing T_L and T_R to unitaries $U_L = [T_L,\, T_L^\perp]$ and $U_R = [T_R,\, T_R^\perp]$ gives the block encoding $W = U_L^* U_R$, with $\alpha = a_{\max}\sqrt{s_r s_c}$.

What "entries computable by a circuit" means. U_R maps e_j to ϕ_j: it must produce the positions of the nonzeros of column j (a column-index oracle). U_L maps e_i to ψ_i: it must produce the positions in row i, and rotate a flag qubit by an angle determined by a_{ij} (a value oracle). For structured matrices, such as stencils on Cartesian grids, these are small arithmetic circuits on indices. For unstructured meshes they need lookup tables, which is the data-access problem noted in the paper.

The constant. $\alpha = a_{\max}\sqrt{s_r s_c}$ is exactly the Schur-test bound for $\|A\|$, so it is never smaller than $\|A\|$, and often noticeably larger. In the random example of Table 3, $\alpha = 15.3$ against $\|A\| = 5.1$. This is the origin of the pessimistic constant C_d in the paper's Proposition 3.8, and of the suggestion to use Kronecker and LCU constructions instead of generic sparse access.

10 Summary of costs

11 How the rules build G in the paper

$G = [T_1 G_1, \ldots, T_L G_L]$ is assembled from the inside out:

1. 1D two-term factors (values and derivatives at Gauss points): LCUs of shifts with diagonal weights, α of the order of their norm.
2. d-dimensional factors: Kronecker products of 1D factors, so the α's multiply. They match the norms because the norms of Kronecker products also multiply.

rule	realization	extra qubits	subnormalization
single A (dense)	Halmos dilation	1	$\|A\|$
AB	$\widetilde{U}_A\widetilde{U}_B$, fresh ancillas	$a_A + a_B$	$\alpha_A\alpha_B$
$A \otimes B$	$U_A \otimes U_B$, permuted	$a_A + a_B$	$\alpha_A\alpha_B$
$\sum_j c_j A_j$	$(V^* \otimes I)\,\mathrm{SEL}\,(V \otimes I)$	$\log_2 k + \max a_j$	$\sum_j c_j\alpha_j$
$[A_0; \ldots; A_{k-1}]$	$\mathrm{SEL}\,(V \otimes I)$	$\log_2 k + \max a_j$	$(\sum_j \alpha_j^2)^{1/2}$
$[A_0, \ldots, A_{k-1}]$	$(V^* \otimes I)\,\mathrm{SEL}$	$\log_2 k + \max a_j$	$(\sum_j \alpha_j^2)^{1/2} \le \sqrt{k}\max\alpha_j$
sparse A	$U_L^* U_R$	$\log_2 n + 1$	$a_{\max}\sqrt{s_r s_c}$
isometry T	$[T,\, T^\perp]$	as needed	1

Table 1: The combination rules. SEL needs *controlled* versions of the U_j, which typically cost a small constant factor more gates.

3. G_ℓ: a stack of the value factor and the d gradient factors, weighted by ω_ℓ and $\beta^{1/4}$, with $\alpha(G_\ell) = (\sum_c \alpha_c^2)^{1/2}$.
4. $T_\ell G_\ell$: a product with an isometry, which costs nothing in α.
5. G: a concatenation over the levels, which costs $\sqrt{L}$.

The only places where α can exceed the norm are step 1 (if the LCU is not tight), step 3 (stacking is tight only if all components are equally saturated) and step 5 (the $\sqrt{L}$). Measuring α_G for the explicit construction (to-do item 1.1) quantifies exactly these three losses.

12 Amplitude amplification: a rotation in a plane

A block encoding produces the wanted vector only as one block of a longer vector. Postselection keeps that block, but succeeds only with some probability. Amplitude amplification raises this probability to nearly one, at a cost proportional to $1/a$ instead of $1/a^2$, where a is the success amplitude. It is the source of the factor κ_α^2 in plain QSVT inversion (Section 2 of the paper).

The setting. Let W be a unitary (e.g. the QSVT circuit for $p(\mathcal{B}/\alpha)$), b a unit vector, and $u = e_0 \otimes b$. Let Π be the orthogonal projector onto the "ancilla $= 0$" subspace, i.e. onto the first block. Then

$$\psi := Wu = \begin{bmatrix} x \\ r \end{bmatrix}, \qquad x = \Pi\psi \text{ (e.g. } x = \sigma_{\min}\mathcal{B}^+ b), \qquad a := \|x\| \le 1.$$

Measuring the ancilla returns $x/\|x\|$ with probability a^2. Repeating until success costs about $1/a^2$ applications of W.

Two reflections. Define

$$S_\Pi = I - 2\Pi, \qquad S_\psi = I - 2\psi\psi^* = W\,(I - 2uu^*)\,W^*, \qquad Q := -S_\psi S_\Pi.$$

S_Π flips the sign of the first block, which is a cheap circuit. S_ψ needs one W^*, one reflection about u (which requires running the preparation of b backwards and forwards), and one W.

The invariant plane. Let $g = \Pi\psi/a$ (the good direction) and $h = (I-\Pi)\psi/\|(I-\Pi)\psi\|$ (the bad direction). They are orthonormal, and with $a = \sin\theta$,

$$\psi = \sin\theta\, g + \cos\theta\, h.$$

Both reflections map $\operatorname{span}\{g, h\}$ into itself: $S_\Pi g = -g$, $S_\Pi h = h$, and S_ψ reflects about the line orthogonal to ψ. In the basis (g, h),

$$S_\Pi = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}, \qquad -S_\psi = 2\psi\psi^* - I = \begin{bmatrix} -\cos 2\theta & \sin 2\theta \\ \sin 2\theta & \cos 2\theta \end{bmatrix}, \qquad Q = \begin{bmatrix} \cos 2\theta & \sin 2\theta \\ -\sin 2\theta & \cos 2\theta \end{bmatrix},$$

so Q is the rotation by 2θ towards g. Hence

$$Q^k \psi = \sin\big((2k+1)\theta\big)\, g + \cos\big((2k+1)\theta\big)\, h.$$

With $k \approx \pi/(4\theta) - \frac{1}{2}$ the coefficient of g is close to one, and a single measurement succeeds with probability $\sin^2((2k+1)\theta) \geq 1 - \sin^2\theta$ for the best integer k. The number of rounds is $k \approx \pi/(4a)$. Each round costs one W and one W^*.

Polynomial view. Since $a = \sin\theta = \cos(\pi/2 - \theta)$,

$$\sin\big((2k+1)\theta\big) = (-1)^k T_{2k+1}(a),$$

with T_n the Chebyshev polynomial of the first kind. Amplitude amplification is therefore QSVT applied to the 1×1 "matrix" a (the singular value of $\Pi W \Pi$ on u), with an odd polynomial of degree $2k+1$ that maps a close to ± 1. Its degree $O(1/a)$ follows the same law as the degree $O(1/\delta)$ of the polynomial approximating δ/t on $[\delta, 1]$.

Application to QSVT inversion. The polynomial acts on $\mathcal{B}/\alpha$, so $p(\mathcal{B}/\alpha) \approx \delta(\mathcal{B}/\alpha)^+ = \sigma_{\min}\mathcal{B}^+$ with $\delta = \sigma_{\min}/\alpha$. For $x = \sigma_{\min}\mathcal{B}^+ b$ and $b \in \operatorname{range}\mathcal{B}$,

$$a = \sigma_{\min}\|\mathcal{B}^+ b\| \;\geq\; \frac{\sigma_{\min}}{\|\mathcal{B}\|} \;\geq\; \frac{\sigma_{\min}}{\alpha} = \frac{1}{\kappa_\alpha}.$$

So at most about $(\pi/4)\kappa_\alpha$ rounds are needed, each containing W of degree $D = O(\kappa_\alpha \log(\kappa_\alpha/\varepsilon))$: in total $O(\kappa_\alpha^2 \log(\kappa_\alpha/\varepsilon))$ calls to the block encoding of $\mathcal{B}$. The two factors κ_α have different origins: one is the degree of p, the other the size of $\sigma_{\min}\mathcal{B}^+ b$. Variable-time amplitude amplification (Childs, Kothari, Somma 2017) and the discrete adiabatic method (Costa et al. 2022) avoid multiplying them and reach $O(\kappa_\alpha \log(1/\varepsilon))$.

Practical points.

- The optimal k depends on a, which is unknown. Exponential search over k (Brassard, Høyer, Mosca, Tapp 2002) or fixed-point amplitude amplification (Yoder, Low, Chuang 2014) cost only a constant factor more.
- Too many rounds rotate past g; the success probability oscillates in k. This is the price of a non-adaptive polynomial, as for QSVT itself.
- S_ψ requires that b can be prepared by a circuit and that this circuit can be inverted. This is a standing assumption of QLSAs.

13 Removing the mass-matrix floor

Where the floor comes from. The multilevel factor of the paper scales each level by a scalar, $\omega_\ell^2 = \theta_\ell/\delta_\ell$ with δ_ℓ the (constant) diagonal of $A_{\beta,\ell} = M_\ell + \sqrt{\beta}K_\ell$. On a mass-dominated level this treats M_ℓ as a multiple of the identity, but $M_\ell = M_1^{\otimes d}$ has eigenvalues in $((h/3)^d, h^d)$, so $\kappa(\delta_\ell^{-1} M_\ell) \to 3^d$. When the finest grid does not resolve β, $\kappa(CA_\beta)$ therefore approaches $3^d/(1 - 4^{-s})$: 4, 12 and 36 for $d = 1, 2, 3$ and $s = 1$. This is harmless for Poisson control on fine grids, but in the time-periodic heat problem all high frequencies are mass-dominated once $\tau \lesssim 2h_L^2$, and the worst frequency sets κ_α.

A better level scaling. Replace the scalar δ_ℓ^{-1} by

$$D_\ell = (M_\ell + c_\ell I)^{-1}, \qquad c_\ell = \sqrt{\beta}\ \mathrm{diag}(K_\ell) \quad \text{(a scalar)},$$

i.e. the exact mass matrix plus Jacobi scaling of the stiffness part, with $C = \sum_\ell \tilde{\theta}_\ell P_\ell D_\ell P_\ell^T$. On mass-dominated levels $D_\ell \approx M_\ell^{-1}$, on stiffness-dominated levels $D_\ell \approx c_\ell^{-1} I$ as before. In the unresolved limit $CM = \sum_\ell \tilde{\theta}_\ell Q_\ell$ with $Q_\ell = P_\ell M_\ell^{-1} P_\ell^T M$ the matrix of the L^2 projection onto V_ℓ ($Q_L = I$), so $\mathrm{spec}(CM) \subset [1, \sum_\ell \tilde{\theta}_\ell]$ and the floor drops to $1/(1-4^{-s}) = 4/3$. In the proof of the paper's Theorem 3.5 the nodal-versus-L^2 comparison (the factor $3d \cdot 3^d$) becomes an identity on mass-dominated levels. Two simpler variants are worse: M_ℓ^{-1} on *all* levels loses on resolved grids (stiffness-dominated levels want a scalar), and switching between M_ℓ^{-1} and Jacobi by level type gives $\kappa \approx 10$ at intermediate β.

The encoding challenge. The new level factor is

$$G_\ell = \sqrt{\tilde{\theta}_\ell} \begin{bmatrix} F_{\mathrm{val}} \\ \beta^{1/4} F_\nabla \end{bmatrix} D_\ell^{1/2}, \qquad F_{\mathrm{val}}^T F_{\mathrm{val}} = M_\ell, \quad F_\nabla^T F_\nabla = K_\ell.$$

M_ℓ, K_ℓ and D_ℓ are all diagonalized by the d-dimensional sine transform, so everything can be read off mode by mode, with μ and k the eigenvalues of M_ℓ and K_ℓ.

- *Value rows: (W2) fails for the plain product.* F_{val} is largest on smooth modes ($\mu \approx h^d$), $D_\ell^{1/2}$ on oscillatory ones ($\mu \approx (h/3)^d$). On a mass-dominated level $\|F_{\mathrm{val}}\|\,\|D_\ell^{1/2}\| \approx 3^{d/2}$, whereas $\|F_{\mathrm{val}} D_\ell^{1/2}\| \le 1$. Encoding the product of the two factors therefore inflates α by $3^{d/2}$ and κ_α by 3^d: the floor moves from $\kappa(CA_\beta)$ into the subnormalization.

- *Gradient rows: nearly aligned.* F_∇ and $D_\ell^{1/2}$ are both largest on oscillatory modes; the loss is a bounded factor.

The remedy: factor through an isometry. Since $F_{\mathrm{val}}^T F_{\mathrm{val}} = M_\ell$ exactly, $U_{\mathrm{val}} := F_{\mathrm{val}} M_\ell^{-1/2}$ is an isometry, and

$$F_{\mathrm{val}} D_\ell^{1/2} = U_{\mathrm{val}}\, g(M_\ell), \qquad g(\mu) = \sqrt{\mu/(\mu + c_\ell)} \in [0, 1].$$

A product with an isometry is always tight. Both factors can be encoded with α equal to their norm, using the construction of Yang, Yu and Zhang (arXiv:2610.06987, Secs. 3.4–3.5):

1. *Functions of* M_ℓ (here $g(M_\ell)$, there $M_\ell^{-1/2}$) are applied in the sine basis: a sine transform (via the QFT), a rotation of a flag qubit by an angle computed from the mode index, so that the amplitude on the flag state 0 is $g(\mu_j)/\max g$, and the inverse transform. This is a block encoding with $\alpha = \max g$, i.e. tight; no polynomial approximation is needed. For D_ℓ the angle depends on all d mode indices, since $M_\ell + c_\ell I$ is not a Kronecker product; the arithmetic is still cheap.

2. *The isometry* U_{val} is the Kronecker product of d one-dimensional isometries $E_1 M_1^{-1/2}$ (E_1: values at the two Gauss points). The product encoding of E_1 and $M_1^{-1/2}$ has $\alpha = \sqrt{4h/3}\,\sqrt{3/h} = 2$, i.e. its block is $F = \frac{1}{2} E_1 M_1^{-1/2}$ with $F^*F = \frac{1}{4} I$. One step of *oblivious* amplitude amplification (Section 12, with the reflection about the input replaced by the reflection about the ancilla state 0) has block $3F - 4FF^*F = 2F$, i.e. it encodes the isometry with $\alpha = 1$, using three calls to U or U^*. This works because F^*F is a multiple of the identity: all singular values are rotated by the same angle. The d-dimensional U_{val} then has $\alpha = 1^d = 1$.

3. *Gradient rows.* The same idea works per direction q: with $E_1 = U_1 M_1^{1/2}$,

$$F_\nabla^{(q)} D_\ell^{1/2} = \big[U_1 \otimes \cdots \otimes C_1 \otimes \cdots \otimes U_1\big] \cdot \Big[\big(M_1^{1/2} \otimes \cdots \otimes I \otimes \cdots \otimes M_1^{1/2}\big) D_\ell^{1/2}\Big],$$

where C_1 (derivatives, $\alpha = \|C_1\| = 2/\sqrt{h}$) sits in position q. The first factor is a Kronecker product of isometries and C_1, hence tight; the second is diagonal in the sine basis, hence tight. The two are nearly aligned (both favour modes that oscillate in direction q), so the product loses only a bounded factor. The directions are stacked.

Measured effect. Table 2 compares, with the same one-dimensional primitives for all three: the paper's Jacobi scaling, the Yang–Yu–Zhang factor ($M_\ell^{-1/2}$ on every level, weights $(1+t_\ell)^{-1}$, i.e. $s = 0$), and the hybrid scaling D_ℓ with $s = 1$. The subnormalization is $\alpha_G^2 = \sum_\ell \alpha_\ell^2$ (concatenation) and $\kappa_\alpha = \alpha_G^2/\sigma_{\min}(\mathcal{B})$.

			$\kappa(CA_\beta)$			$\kappa_\alpha = \alpha_G^2/\sigma_{\min}(\mathcal{B})$			
d	L	β	Jacobi	YYZ	hybrid	Jacobi, raw	Jacobi, tight	YYZ	hybrid, tight
1	8	1	5.0	7.3	5.0	16.0	16.0	35.7	16.0
1	8	10^{-4}	5.4	7.2	6.2	17.2	16.7	35.4	18.3
1	8	10^{-8}	3.6	4.4	4.5	7.8	7.4	17.1	8.3
1	8	10^{-12}	2.3	7.0	1.3	5.0	3.9	9.3	2.1
2	5	1	4.1	9.8	4.1	26.6	18.0	37.8	18.1
2	5	10^{-4}	2.9	8.9	3.6	19.0	13.6	29.8	15.6
2	5	10^{-8}	3.8	4.1	1.3	12.9	8.0	8.3	3.6
2	5	10^{-12}	11.5	5.0	1.3	21.2	11.8	5.0	1.4
3	3	1	4.2	9.7	4.1	43.2	17.8	30.2	17.8
3	3	10^{-4}	2.8	6.9	2.4	22.1	10.5	16.0	8.6
3	3	10^{-8}	20.5	3.0	1.5	69.8	25.8	3.3	2.0
3	3	10^{-12}	24.4	3.0	1.3	67.4	24.5	3.0	1.3

Table 2: Preconditioner quality and effective condition number for three level scalings. "Raw": the one-dimensional value factor E_1 encoded directly ($\alpha = \sqrt{4h/3}$ against $\|E_1\| = \sqrt{h}$). "Tight": $E_1 = U_1 M_1^{1/2}$ with the isometry U_1 and a diagonal factor in the sine basis. α_G is computed from the combination rules with these primitive subnormalizations (`qp_mass_floor.py`).

Three observations.

- *The remedy works.* In the unresolved 3D cases it reduces κ_α from about 25 (Jacobi, tight) to 1.3. On resolved grids it is within about 15% of the Jacobi route, sometimes slightly better, sometimes slightly worse. The floor does not reappear in α.
- *Tight primitives help the paper's construction too.* Even with Jacobi scaling, encoding the value factor through the isometry U_1 lowers κ_α by up to a factor 2.4 in 3D (e.g. $43 \to 18$ at $\beta = 1$), because the raw value factor loses $(2/\sqrt{3})^d$ in α^2 per level and the raw gradient factor more. This is relevant for Proposition 3.8 and to-do item 1.1.
- *Yang–Yu–Zhang* encode tightly but do not damp mass-dominated levels and use M_ℓ^{-1} also on stiffness-dominated levels; their $\kappa(CA_\beta)$ is up to 9.8 on resolved grids, and their κ_α is about twice ours in 1D and 2D.

What remains.

- The gate cost per call to G: per level a sine transform and its inverse (QFT on $\ell+1$ qubits per dimension), reversible arithmetic for the rotation angle, and the three-call amplification

of the value isometry. All of this is polylogarithmic, but it multiplies the cost of every call to G by a constant that has to be weighed against the gain in κ_α.

- Adapting the robustness proof to D_ℓ (straightforward: the nodal lemma becomes an identity on mass-dominated levels).
- Prior art: the encoding of $M_\ell^{-1/2}$ and of the normalized value map is due to Yang–Yu–Zhang. New would be the hybrid scaling D_ℓ with damped weights, which is what makes the floor disappear without losing on resolved grids.

14 Numerical verification

construction	$\|W^*W - I\|_{\max}$	error of the selected block
single matrix (Halmos)	$1.2 \cdot 10^{-15}$	0
product, fresh ancillas	$1.9 \cdot 10^{-15}$	$8.4 \cdot 10^{-17}$
product, *same* ancilla	–	$7.9 \cdot 10^{-1}$ (fails, as predicted)
Kronecker product	$1.3 \cdot 10^{-15}$	$5.6 \cdot 10^{-17}$
LCU, $k = 4$ complex coefficients	$3.3 \cdot 10^{-15}$	$3.9 \cdot 10^{-16}$
stacking, $k = 4$	$1.3 \cdot 10^{-15}$	$3.9 \cdot 10^{-17}$
concatenation, $k = 4$	$1.6 \cdot 10^{-15}$	$3.9 \cdot 10^{-17}$
sparse, $n = 8$, $s_r = s_c = 4$	$1.7 \cdot 10^{-15}$	0
amplitude amplification, $a = 0.23$, 3 rounds	–	$2.5 \cdot 10^{-15}$ (rotation formula)

Table 3: Dense checks of all constructions (`qp_block_encoding_rules.py`, random complex matrices, $n = 4$ unless stated).